VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY

A COURSE-BASED PROJECT REPORT ON

JOB PORTAL MANAGEMENT SYSTEM

A Desktop-Based Recruitment and Job Application Platform

Submitted to the

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

in partial fulfilment of the requirements for
Object Oriented Programming Through JAVA Laboratory (25PC2CS201)

BACHELOR OF TECHNOLOGY
IN COMPUTER SCIENCE AND ENGINEERING

SUBMITTED BY

STUDENT NAME	ROLL NUMBER
NAVEEN	25071A05H4
JATHIN	25071A05F5
ANWESH	26075A0517
PARVEZ KUMAR	25071A05J4

Under the guidance of

Dr. P. Anjusha

(Course Instructor)
Assistant Professor, Department of CSE, VNRVJIET

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY
Vignana Jyothi Nagar, Pragathi Nagar, Nizampet, Hyderabad – 500 090, Telangana, India

OCTOBER 2026

VALLURUPALLI NAGESWARA RAO VIGNANA JYOTHI INSTITUTE OF ENGINEERING & TECHNOLOGY

An Autonomous Institute, NAAC Accredited with 'A++' Grade, NBA Accredited for CE, EEE, ME, ECE, CSE, EIE, IT B.Tech Courses, Approved by AICTE, New Delhi, Affiliated to JNTUH
Vignana Jyothi Nagar, Pragathi Nagar, Nizampet (SO), Hyderabad – 500090, TS, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the course-based project report entitled "Job Portal Management System: A Desktop-Based Recruitment and Job Application Platform" is a bonafide work carried out by the students listed below, in partial fulfilment of the requirements for the Object Oriented Programming Through JAVA Laboratory (25PC2CS201) during the academic year 2026–2027.

- NAVEEN (25071A05H4)
- JATHIN (25071A05F5)
- ANWESH (26075A0517)
- PARVEZ KUMAR (25071A05J4)

The work presented in this report is based on the project implementation carried out by the team.

Course Instructor	Head of the Department
Dr. P. Anjusha	Dr. V. Baby
Assistant Professor, Department of CSE	Department of CSE

DECLARATION

We hereby declare that the course-based project work entitled “Job Portal Management System: A Desktop-Based Recruitment and Job Application Platform” is our own work carried out as part of the Object Oriented Programming Through JAVA Laboratory. This report presents the system, database design, implementation flow and testing outcomes of our project.

Place: Hyderabad

Date: October 2026

NAVEEN — 25071A05H4

JATHIN — 25071A05F5

ANWESH — 26075A0517

PARVEZ KUMAR — 25071A05J4

ACKNOWLEDGEMENT

We express our sincere gratitude to the management and Principal of VNR Vignana Jyothi Institute of Engineering & Technology for providing the facilities required to complete this project.

We thank Dr. P. Anjusha, Assistant Professor, Department of CSE, for her guidance and support. We also thank Dr. V. Baby, Head of the Department of CSE, for the encouragement and academic support provided.

Finally, we thank our teammates and everyone who contributed to the successful completion of the Job Portal Management System.

TABLE OF CONTENTS

SECTION	CONTENTS
1	Abstract
2	Introduction and Objectives
3	System Design and Database
4	Implementation and Testing
5	Conclusion
6	References

1. ABSTRACT

The Job Portal Management System is a desktop-based application that connects job seekers with employers on a single platform. Built using Java 21, Java Swing, JDBC and MySQL, it allows job seekers to register, build a profile, search and filter job openings and apply online, while employers and administrators can post jobs, review applicants and manage the shortlisting process. User accounts, job listings, applications and hiring status are stored in a relational database.

2. INTRODUCTION AND OBJECTIVES

Finding a suitable job and hiring the right candidate are both time-consuming when done manually. Candidates struggle to track openings and applications, while employers struggle to organise resumes and responses. The Job Portal Management System provides a structured, centralised solution: job seekers sign in, browse available openings, apply with their profile details and track the status of each application, while employers post vacancies and update applicant status.

The main objectives of the project are:

- Provide registration, login and logout for three user roles: Job Seeker, Employer and Administrator.
- Allow employers to post, edit and close job vacancies, organised by category (IT & Software; Engineering; Finance & Marketing; Other Industries).
- Allow job seekers to create and update a profile, search and filter jobs by title, location, category and salary, and apply to openings.
- Let employers view applicants, and shortlist, accept or reject applications, with status recorded for every application.
- Enable administrators to manage users and job listings and to view summary reports of jobs and applications.

3. SYSTEM DESIGN AND DATABASE

The system follows a three-layer design. The Java Swing presentation layer displays login and registration forms, job listings, application forms, employer dashboards and admin screens. The application-logic layer manages authentication, role-based access, job posting, application processing and status updates. JDBC connects the Java application to the MySQL database.

Technologies used: Java 21, Java Swing, JDBC, MySQL, Maven, GitHub and MySQL Workbench.

Core database tables:

- **users** — account details, role (job seeker, employer or admin) and login-related information.
- **job_seekers** — profile details such as qualification, skills, experience and resume information.
- **employers** — company name, industry, contact and location details.
- **jobs** — job title, description, location, salary, category, posting date and the employer who posted it.
- **applications** — which job seeker applied to which job, application date and current status (Applied, Shortlisted, Rejected, Selected).
- **categories** — job categories used to organise and filter openings.

The project uses INSERT to store new users, jobs and applications, SELECT with WHERE to retrieve user-specific and filtered job data, UPDATE to modify profiles, job details and application status, DELETE to remove expired or closed listings, ORDER BY to sort jobs by date or salary, GROUP BY with COUNT to aggregate the number of applicants per job, and JOIN to combine users, jobs and applications. JDBC PreparedStatement supports parameterised database operations and helps guard against SQL injection.

4. IMPLEMENTATION AND TESTING

Main flow: registration/login → role-based dashboard → (job seeker) search jobs → view details → apply → track application status; (employer) post a job → view applicants → shortlist, select or reject; (admin) manage users and listings → view reports.

Core test cases and outcomes:

Test Case	Result
Registration, login and logout for all three roles	Passed
Employer posting, editing and closing a job	Passed
Job search and filtering by title, location, category and salary	Passed
Job seeker applying to a job (duplicate applications prevented)	Passed
Employer updating application status and job seeker viewing it	Passed
Admin user management and applicant-count report	Passed

5. CONCLUSION

The Job Portal Management System simplifies recruitment by bringing job seekers, employers and administrators onto one platform and by making the application process transparent and trackable. The project demonstrates desktop UI development with Swing,

database integration through JDBC, relational data management in MySQL and role-based access control. Future improvements could include resume upload and parsing, email notifications, job recommendations based on skills, interview scheduling, a web or mobile version and richer analytics for employers.

6. REFERENCES

- Oracle Java documentation: https://docs.oracle.com/en/java/
- MySQL Reference Manual: https://dev.mysql.com/doc/
- Apache Maven documentation: https://maven.apache.org/guides/